

Tinnaphop Sonsang

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PROFESSIONAL SUMMARY

Computer Engineering and AI graduate focused on applied AI, with hands-on experience building computer vision and data-driven solutions. Experienced in taking projects from data preparation and experimentation to practical deployment.

EDUCATION

Bachelor of Engineering – Computer Engineering & Artificial Intelligence **2022-2026**
Walailak University, GPAX: 3.65 (First-class Honors, Graduation: 7 May 2026)

CERTIFICATIONS

- **EF SET English Certificate** ([View certificate: cert.efset.org/en/gCMcLJ](https://cert.efset.org/en/gCMcLJ))
Overall score: 56/100 - B2 Upper Intermediate (CEFR); Reading: C1 Advanced (64/100)
- **Microsoft Learn: Azure Certificate - Operationalize machine learning models (MLOps)**
ML pipelines, experiment tracking, model deployment, CI/CD, and monitoring

PROFESSIONAL EXPERIENCE

AI & Computer Vision - Intern **2025-2026**
Thaidotrun Co., Ltd. (Merged into ASICS), Bangkok

Project: Automated Runner Highlight Extraction System

- Developed an automated pipeline for detecting, tracking, and extracting individual runner highlights from crowded race footage, with generated clips typically lasting 10-20 seconds for social media use.
- Evaluated object detection and tracking approaches, including CNN and Transformer-based models, and applied OCR with spatial weighting to identify runners and improve highlight selection.
- Achieved ~60% average success rate in automated highlight generation. The remaining failures were mainly caused by tracking errors, resulting in clips that were too short or excessively long

Project: Shoe Detection & Classification Pipeline for Race Insights

- Developed a computer vision pipeline for shoe-brand recognition from race images and video. Initially explored object detection, but due to available resources and project timeline, evaluated an open-source shoe detection model and adopted a detection-and-crop approach followed by a custom classification model.
- Built the dataset and training workflow by removing redundant video frames using DINO-based feature extraction, HDBSCAN clustering, and color similarity, followed by representative image selection, human verification, pseudo-labeling, and data augmentation. Iterated through limited-data training, zero-shot model evaluation, and supervised training as additional data became available, with experiments managed through Azure ML.
- Improved test-set performance from 0.74 to 0.85 ± 0.05 through iterative data and model development. Processed 150,000+ images down to ~35,000 verified images and reduced dataset preparation and annotation time from 7-10 days to 1-2 days. Performance was strong on the three major shoe brands, while long-tail brands remained less consistent.

Data Scientist - AI & Research (Short-Term Contract) **2026**
Lief Capital Asset Management, Bangkok

- Worked with an existing AI and data processing codebase on Azure Machine Learning, gaining hands-on experience across data gathering, EDA, preprocessing, feature engineering, model training, and evaluation.
- Analyzed large-scale, multi-decade financial data workflows and improved data processing and external data ingestion, using parallel processing, connection reuse, vectorized computation, and Polars to reduce runtime from over 1 hour to under 30 minutes.
- Contributed to a machine learning-based financial ranking pipeline, working with feature development, model optimization, ranking evaluation, and time-aware backtesting using Walk-Forward Validation.

- Explored maize tassel counting in dense UAV imagery, where identifying individual tassels can become difficult due to high density and overlap. Investigated spatial density estimation as an alternative approach for more robust counting.
- Developed and evaluated a deep learning model for spatial density estimation and object counting, including model fine-tuning, data augmentation, training optimization, and multi-seed testing.
- Evaluated model reliability through uncertainty estimation, density-stratified performance analysis, visual attribution, occlusion sensitivity, and failure-mode analysis to understand model behavior across different image conditions.
- Achieved 34.0 ± 3.3 MAE across 5 random seeds on an internal held-out test set. The next phase will involve collecting real-world UAV data from my family's maize field during the next planting cycle to further evaluate the model for crop monitoring and precision agriculture.

Thesis: Detection of White Blood Cells in Raptors**2024-2025***Walailak University, Graduation Thesis - Bachelor of Engineering in Computer Engineering and Artificial Intelligence*

- Developed a deep learning object detection system for identifying six blood-cell classes in raptor blood smear images using 1,180 manually annotated images
- Benchmarked YOLO-11l, YOLO-12l, YOLO-26l, and RT-DETR-l across different augmentation strategies, achieving the best baseline with YOLO-26l at 0.957 mAP@0.5, 0.746 mAP@0.5:0.95, and 93.4% recall. Used Grad-CAM to analyze model behavior and augmentation effects.
- Built a full-stack AI inference application using Next.js, React, and PostgreSQL (Supabase), deploying the web application on Vercel and running the trained model directly in the browser using WebAssembly (WASM) for client-side inference and batch image processing.
- Conducted the research in collaboration with faculty advisors from Walailak University's School of Engineering and Technology and Akkharatchakumari Veterinary College, and authored a manuscript based on the study for submission to an international journal.

Technical Skills**Artificial Intelligence**

- **Learning:** Supervised Learning, Unsupervised Learning
 - Supervised: Machine Learning, Deep Learning, Classification, Regression
 - Unsupervised: Clustering, Representation Learning
- **AI Methods:** Ensemble Learning, Few-Shot Learning, Pseudo-Labeling, Feature Engineering, Model Evaluation, Model Benchmarking
- **Computer Vision:** Object Detection, Object Tracking, Image Classification, Pose Estimation, OCR, Image Processing, Data Augmentation, Grad-CAM, Explainable AI
- **Data & AI Workflow:** Data Processing, Data Preprocessing, Dataset Preparation, Dataset Deduplication, Data Annotation, Data Quality, ETL, Parallel Processing, Workflow Automation
- **Models & Tools:** CNN, Transformers, Vision Transformers, YOLO, RT-DETR, Grounding DINO, XGBoost, Scikit-learn, PyTorch, OpenCV, HDBSCAN, Pandas, NumPy, Polars, PySpark, Azure ML
- **NLP & Applied AI:** Natural Language Processing, Tokenization, SentencePiece, LLM API Integration, Time-Series Modeling, IoT Integration

Web Development

- **Frontend:** JavaScript, TypeScript, Next.js, React, Astro
- **Backend & Database:** REST APIs, PostgreSQL, Supabase, SQL
- **Engineering & Deployment:** Docker, Git, GitHub, Microsoft Azure, Model Deployment

Language

- Thai: Native
- English: Upper-Intermediate (B2)